EXPERIMENT 7: Phasor Relationship for Circuit Element

INTRODUCTION

Phase

The phase of an AC is meant by the fraction of the time period that has been elapsed since the current last passed the zero position of reference.

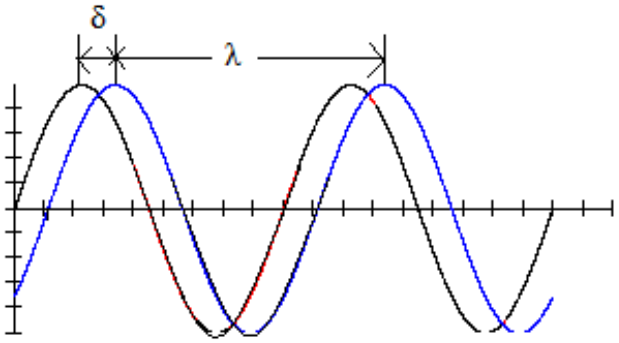


Figure 1

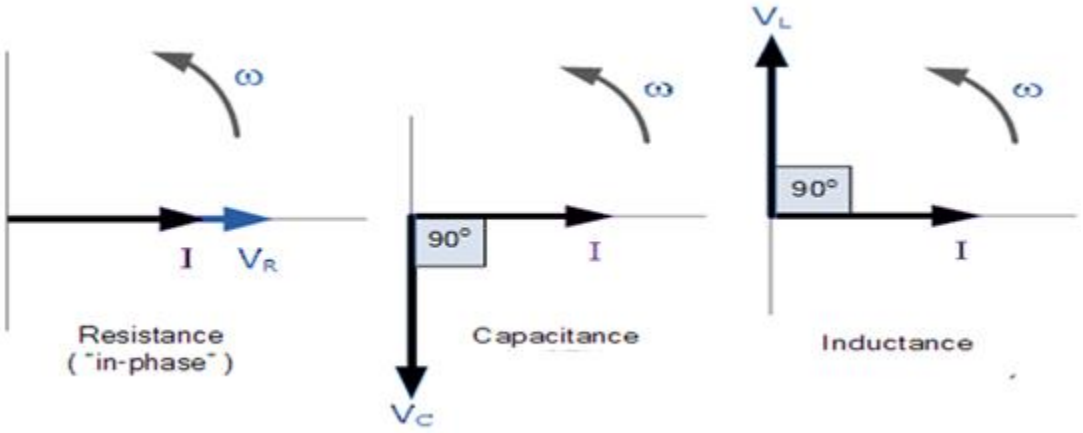


Figure 2: Phase angle diagram

The difference of phase of two signals is expressed by

$$\theta = \frac{\delta}{\lambda} * 360^{\circ}$$

Where, δ is the distance between zero or peak value of the two signals and λ is the wavelength.

OBJECTIVE

1. To measure the phase difference between voltage and current in R-R circuit and to observe phase relation with frequency in R- R circuit.
2. To measure the phase difference between voltage and current in R-C circuit and to observe phase relation with frequency in R- C circuit.
3. To measure the phase difference between voltage and current in R-L circuit and to observe phase relation with frequency in R- L circuit.

PART A: MEASUREMENT OF PHASE SHIFT IN R-R CIRCUIT.

EQUIPMENT AND COMPONENT

1. Oscilloscope with two probe
2. Function Generator
3. Tray A1: Resistor R04 (1kΩ), Resistor 01 (100Ω)
4. BNC to Alligator Clip

PROCEDURE

1. Wire up the circuit as shown in Figure 3. Connect the AFG to AC power and turn ON the AFG. Select the sinusoidal signal, set the frequency to 500Hz and amplitude to 0.5V

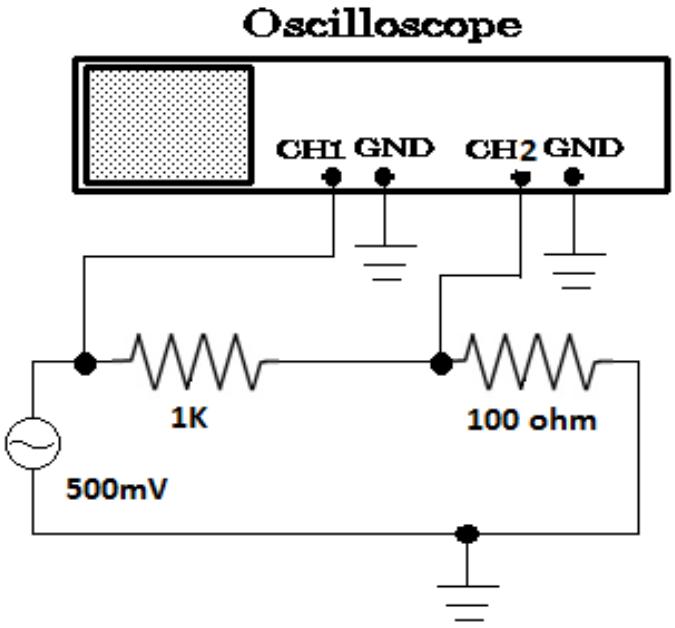


Figure 3: R-R circuit

2. Vary the frequency and measure the distance between zero or peak value of the two signals, δ and λ,wavelength. Record your result in Table 1
3. Calculate the phase difference.

RESULT

Table 1

Input voltage: 500mV (AC)

SL	Frequency (Hz)	Distance between zero or peak value of the two signals, δ	λ, wavelength	Phase Difference(Θ) $\Theta = \frac{\delta}{\lambda} * 360^{\circ}$
1	500			
2	1k			
3	5k			
4	10k			

PART B: MEASUREMENT OF PHASE SHIFT IN R-C CIRCUIT.

EQUIPMENT AND COMPONENT

1. Oscilloscope with two probe
2. Tray A1: Resistor R01 (100Ω)
3. Tray A2: Capacitor C04 (1uF)
4. Function Generator
5. Alligator Clip
6. Digital Multimeter

PROCEDURE

1. Wire up the circuit as shown in Figure 4. Connect the AFG to AC power and turn ON the AFG. Select the sinusoidal signal, set the frequency to 500Hz and amplitude to 0.5V

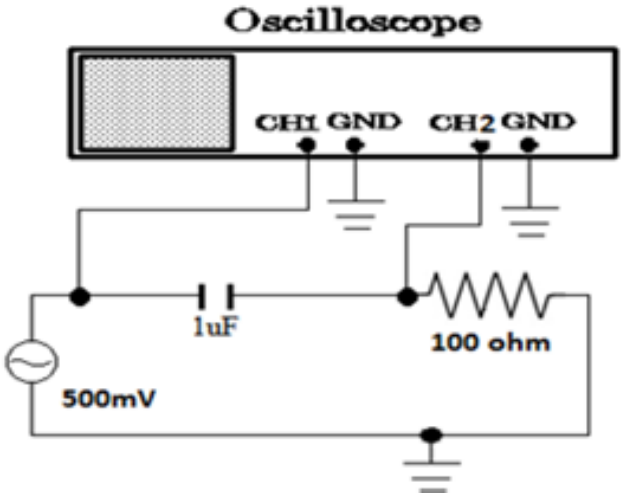


Figure 4: RC Circuit (a)

2. Vary the frequency and measure the distance between zero or peak value of the two signals, δ and λ,wavelength. Calculate the phase difference.
3. By using digital multimeter, measure voltage across resistor and capacitor. Calculate the phase difference.
4. Record your result in Table 2.
5. Wire up the circuit as shown in Figure 5.

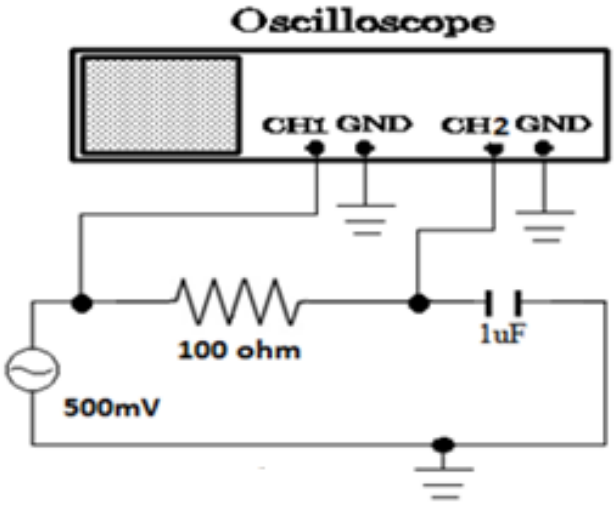


Figure 5: RC Circuit (b)

6. Vary the frequency and measure the distance between zero or peak value of the two signals, δ and λ,wavelength. Calculate the phase difference.
7. By using digital multimeter, measure voltage across resistor and capacitor. Calculate the phase difference.
8. Record your result in Table 3.

RESULT

Table 2

Input voltage: 500mV (AC)

	Frequency (Hz)	δ	λ	V_R	V_C	Phase Difference(Θ), degree	
						$\Theta = \frac{\delta}{\lambda} * 360$	$\Theta = \tan^{-1}\frac{Vc}{V_R}$
1	500						
2	1k						
3	5k						
4	10k						

Based on your observation, is the output waveform Leading or Lagging?

Table 3

Input voltage: 500mV (AC)

	Frequency (Hz)	δ	λ	V_R	V_C	Phase Difference(Θ), degree	
						$\Theta = \frac{\delta}{\lambda} * 360$	$\Theta = \tan^{-1}\frac{V_R}{Vc}$
1	500						
2	1k						
3	5k						
4	10k						

Based on your observation, is the output waveform Leading or Lagging?

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PART C: MEASUREMENT OF PHASE SHIFT IN R-L CIRCUIT.

EQUIPMENT AND COMPONENT

1. Oscilloscope with two probe
2. Tray A1: Resistor R01 (100Ω)
3. Tray A3: Inductor L01 (15mH)
4. Function Generator
5. Alligator Clip
6. Digital Multimeter

PROCEDURE

1. Wire up the circuit as shown in Figure 6. Connect the AFG to AC power and turn ON the AFG. Select the sinusoidal signal, set the frequency to 500Hz and amplitude to 0.5V

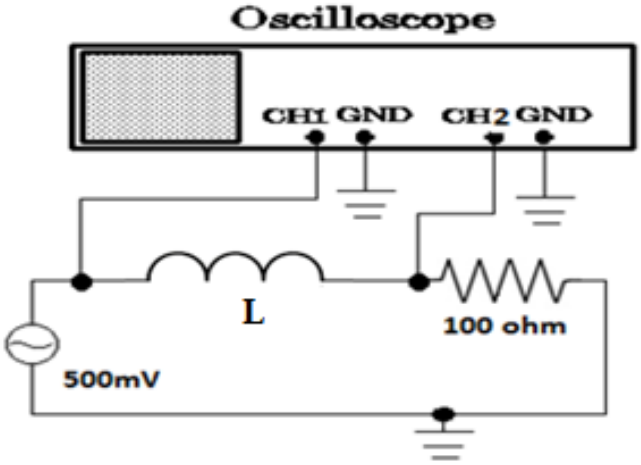


Figure 6: RL Circuit (a)

2. Vary the frequency and measure the distance between zero or peak value of the two signals, δ and λ,wavelength. Calculate the phase difference.
3. By using digital multimeter, measure voltage across resistor and inductor. Calculate the phase difference.
4. Record your result in Table 4.
5. Wire up the circuit as shown in Figure 7.

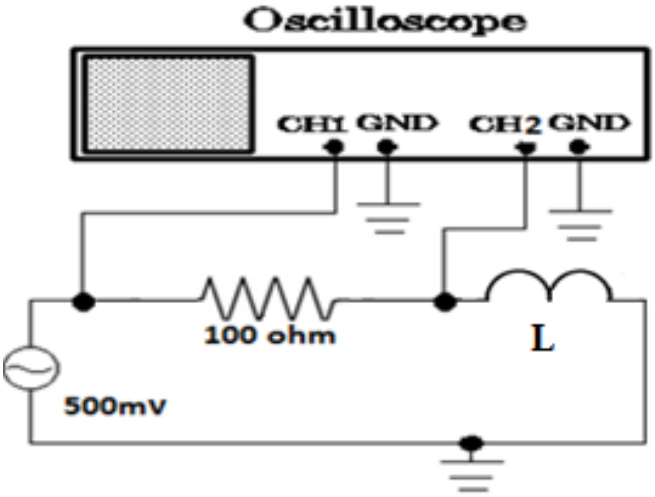


Figure 7: Figure 4: RC Circuit (b)

6. Vary the frequency and measure the distance between zero or peak value of the two signals, δ and λ,wavelength. Calculate the phase difference.
7. By using digital multimeter, measure voltage across resistor and inductor. Calculate the phase difference.
8. Record your result in Table 5.

RESULT

Table 4

Input voltage: 500mV (AC)

	Frequency (Hz)	δ	λ	V_R	V_L	Phase Difference(Θ), degree	
						$\Theta = \frac{\delta}{\lambda} * 360$	$\Theta = \tan^{-1}\frac{V_L}{V_R}$
1	500						
2	1k						
3	5k						
4	10k						

Based on your observation, is the output waveform Leading or Lagging?

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Table 5

Input voltage: 500mV (AC)

	Frequency (Hz)	δ	λ	V_R	V_L	Phase Difference(Θ), degree	
						$\Theta = \frac{\delta}{\lambda} * 360$	$\Theta = \tan^{-1}\frac{V_R}{V_L}$
1	500						
2	1k						
3	5k						
4	10k						

Based on your observation, is the output waveform Leading or Lagging?

DISCUSSION

1. Elaborate your result and findings in this experiment.
2. Show all working calculation for Table 1 to Table 5.